



Prediction of index futures movement using TimeGAN and 3D-CNN: Empirical evidence from Korea and the United States

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HIGHLIGHTS

- We propose an effective algorithmic trading model in the futures market.
- TimeGAN performs data augmentation by incorporating temporal characteristics.
- Augmented data is expanded to 3D and 3DCNN learned the multidimensional structure.
- The model offers optimal investment advice considering performance and cost.

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ABSTRACT

Over the past several years, a growing number of researchers and investors have displayed a keen interest in predicting stock price movements, actively conducting research that leverages cutting-edge deep learning models to capture the complexities of the stock market. In contrast, research on predicting price movements in the futures market, which is essential for risk management and forecasting the future value of underlying assets, is nearly absent. In this study, we propose a robust and efficient algorithmic trading model designed for the futures market, which exhibits significant time-series volatility, similar to the stock market. We assess the model's performance in both the Korean and U.S. markets. Using time-series generative adversarial networks (TimeGAN), we augment time-series data into three dimensions and employ a 3-dimensional convolutional neural network (3DCNN) to identify temporal correlations within a multidimensional spatial structure. We thoroughly assess the performance of the proposed model by subjecting it to rigorous backtesting within a simulated environment that closely emulates real-world conditions in the futures market. The results demonstrate improvements of approximately 1.35 times in risk-adjusted returns and up to 6390 times in cost efficiency over both short- and long-term timeframes. This demonstrates that the TimeGAN augmentation and 3D expansion method outperforms existing methodologies in terms of risk-adjusted returns and cost efficiency, effectively capturing the volatility of the futures market. The main objective of this research is to provide investors with a more proficient decision-making tool through algorithmic trading that seamlessly functions in the real-world futures market.

1. Introduction

Financial markets exert significant influence over a wide range of domains, including business, education, employment, technology, and the overall economy [1]. Over the years, investors and researchers have sought to develop and evaluate models that elucidate the behavior of

stock prices [2]. Nevertheless, analyzing stock market movements and price behaviors is challenging, given the market's volatility, nonlinearity, nonstationarity, and nonparametric nature and the presence of noise [3]. As a result, as artificial intelligence has progressed, deep learning models for time series have been proposed, demonstrating notable efficiency gains compared to traditional statistical models [4–8].

While research in the stock market is thriving, studies predicting

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Nomenclature and Symbols

CNN	Convolutional Neural Network
GAN	Generative Adversarial Networks
TG, RG, LG	TimeGAN, RNN-GAN, LSTM-GAN
3D	3-Dimension
$X_{1:t}$	Embedding layer maps the data
$\tilde{X}_{1:t}$	Recovery layer maps the data
$h_{1:t}$	Latent vector due to the embedding layer
$\hat{h}_{1:t}$	Latent vector due to the Generate layer
$z_{1:t}$	Random noise vector
$\tilde{y}_{1:t}$	prediction value whether real or fake of the discriminate layer
$\mathcal{L}_R$	Reconstruction Loss
$\mathcal{L}_U$	Unsupervised Loss
$\mathcal{L}_S$	Supervised Loss

price movements in the futures market are notably lacking. Futures represent a type of derivative that allows for high-leverage trading based on predictions of underlying asset price fluctuations [9]. Typically, the futures market is more liquid than the stock market and plays an essential role in risk management rather than speculative actions. Thus, predicting movements in the futures market can be even more critical than predicting stock market movement. Cuny [10]. In particular, index futures offer insight into the future value of the stock market and can serve as a means to mitigate risks associated with stock market investment. Therefore, predicting the future value of index futures is paramount. From an investor's perspective, making precise predictions about the price movements of derivatives such as futures enables attaining greater excess returns through leveraged investments compared to other financial products. However, most ongoing research on the futures market is limited to the commodity futures market [11–14]. Thus, there is an evident need for a systematic and objective approach to predict movements in index futures.

As in the stock market, predicting movements in the futures market is challenging due to significant time-series noise and restricted data availability. Unlike the stock market, the futures market has expiration dates, necessitating rollovers to subsequent contracts at each expiration. Consequently, as the rollover period approaches, trading volumes fluctuate sharply. Therefore, developing predictive models that capture and reflect these time-series complexities is essential. While deep learning in other domains, such as images and text, has utilized data augmentation to improve performance, time series prediction, which involves capturing temporal dependencies, presents challenges in regard to applying data augmentation [15].

Considering this background, this research aims to propose an algorithm for predicting the movement of futures prices using the TimeGAN time-series data augmentation technique and to offer investors a more proficient decision-making tool through algorithmic trading that seamlessly functions in the real-world futures market. The contributions of this paper are tow below:

1. The augmented data are expanded into three dimensions, and a 3DCNN model is applied to identify the temporal correlations in a multidimensional spatial structure. This approach enables the effective identification of uncertainties at each point in time, which could not be captured due to the independence issue of samples when augmented in the form of existing samples. In other words, by reflecting various scenarios that can occur at each point in time, it is possible to minimize the uncertainty that arises when predicting with only one figure per time point.

2. We propose a new trading strategy aimed at achieving high returns on investment with low volatility by predicting movements in the futures market. The practicality of the proposed model is evaluated through backtesting on KOSPI 200, S&P 500, and NASDAQ 100 futures markets, comparing it with existing time-series augmentation methods such as RNN-GAN and LSTM-GAN. This evaluation considers all typical conditions found in real futures investments, including margin requirements and contract sizes.

The structure of this paper is as follows: Section 1 describes the background of stock and futures prediction and highlights the uniqueness of this research. Section 2 reviews previous studies, while Section 3 provides a detailed explanation of the methodologies used in this research. Section 4 describes the research procedure. Section 5 presents the experimental results, and finally, Section 6 concludes and suggests future directions.

2. Literature review

2.1. Forecasting stock price

Research aimed at predicting stock prices has long been a subject of interest among researchers. Early studies primarily employed technologies such as rough sets, probabilistic neural networks, and fuzzy techniques [16–23].

Recently, research has progressed in predicting the direction or future prices of the stock market using various machine learning techniques and ensemble methods. Yakup et al. (2011) proposed two models based on artificial neural networks (ANNs) and support vector machines (SVMs) for predicting the movement direction of the Istanbul Stock Exchange (ISE) National 100 Index. They selected ten technical indicators as inputs and measured performance. The experimental results indicated that the ANN model significantly outperformed the SVM model. Basak et al. [24] conducted tests on various companies using the random forest and extreme gradient boosting (XGBoost) models to predict whether stock prices would rise or fall compared to their prices over the preceding n days, reporting enhanced accuracy compared to the existing prediction performance.

Research actively employing machine learning methodologies in conjunction with traditional time series models has also been actively progressing. Zhang [25] proposed a hybrid methodology that leverages the unique strengths of both the autoregressive integrated moving average (ARIMA) model and the artificial neural network (ANN) model to overcome the limitation of ARIMA in easily capturing nonlinear patterns. Pai & Lin [26] introduced a hybrid model that combines the ARIMA model with the support vector machine (SVM) model.

The advancement of deep learning has led to active research in applying deep learning models to predict stock prices, particularly in the context of highly nonlinear financial data, demonstrating superior performance compared to traditional research. Fischer & Krauss [27] conducted a study using long short-term memory (LSTM) networks to predict directional movements for the constituent stocks of the S&P 500. Long et al. [28] introduced a novel end-to-end model called the multi-filter neural network (MFNN), designed for feature extraction and price movement prediction in financial time series samples. Their research surpassed existing models in terms of accuracy, profitability, and stability. Research focused on index futures is as follows: Zhang et al. [29] predicted index futures using a hybrid approach that combined ARIMA-LSTM and Wavelet Transform, demonstrating robust predictive performance even amid the shocks caused by COVID-19. Liang & Jia [30] proposed a model that combines online response data with financial market data to predict the futures market, showing significant improvements in performance.

To date, research focused on implementing new deep learning models in the stock market continues to progress steadily [31–35].

2.2. Data augmentation

Deep learning models require that a significant number of parameters be learned, rendering them dependent on extensive training data. However, many real-world tasks suffer from a shortage of training data. Data augmentation refers to techniques used to address the problem of limited training data by generating new data or expanding the dataset through data transformations. The progress of deep learning models has spurred extensive research in the field of data augmentation.

Data augmentation techniques have primarily been studied in the computer vision domain. [36] compared the performance of various image data augmentation techniques applied to three medical case studies: skin melanoma diagnosis, histopathological image analysis, and breast magnetic resonance imaging scan analysis. [37] conducted benchmarking of different data augmentation methods to determine the most suitable approaches for enhancing the performance of convolutional neural networks (CNNs).

Research on the application of various data augmentation techniques for time series data has also been actively pursued. Lee & Kim [38] introduced a data augmentation and normalization framework called columnwise random shuffling; it is applicable to Nunit and CNN, which are innovative frameworks for successful learning of high-level features from ultrahigh-dimensional time series data for stock market prediction. Semenoglou et al. [39] proposed that incorporating data augmentation techniques, ranging from basic transformations and adjustments to advanced generative models, in combination with a novel upsampling approach, can significantly enhance predictive performance when used alongside deep neural networks.

Recently, there has been research on data augmentation using generative adversarial network (GAN) models, and GANs have also been applied to augment time series data. [40] proposed the MM-GAN model for MRI augmentation, aimed at filling training sets by sampling new data from constrained data distributions. Mogren [41] proposed the RNN-GAN model, which combines RNN and GAN, and demonstrated its superior performance in generating classical music. Yoon et al. [42] were the first to introduce the time-series generative adversarial network (TimeGAN) framework, which preserves temporal dynamics in generated time-series data. TimeGAN combines unsupervised learning flexibility with the control typically offered by supervised training. In efforts to enhance predictive accuracy, Zhang et al. [43] augmented data with TimeGAN and applied it to SVR, XGB, and RF, comparing the results to using only the original data.

3. Methodology

3.1. Time-series generative adversarial networks

Time-series data are characterized by their inherent temporal dependencies and distinctive correlations over time, creating a challenge for conventional generative adversarial networks (GANs) to model effectively. GANs do not inherently possess a built-in understanding of the unique temporal correlations of time-series data. GANs independently process each data point without taking into account the essential sequential dependencies that are often critical for modeling time-series data. To overcome this limitation and proficiently handle time-series data, researchers have developed specialized variations of GANs and alternative deep learning architectures. TimeGAN [42] introduces the use of a stepwise supervised loss in addition to the unsupervised adversarial loss, allowing it to capture stepwise conditional distributions of the data. It employs embedding networks to reduce the high dimensionality of the adversarial learning space. This is the key difference from existing time-series GAN models. The supervised loss enables robust generation by reflecting the time-series data at the next time step, and by generating data in a low-dimensional embedding space, it prevents overfitting caused by noise in the time-series data.

TimeGAN consists of four parts, as shown in Fig. 1. The embedding

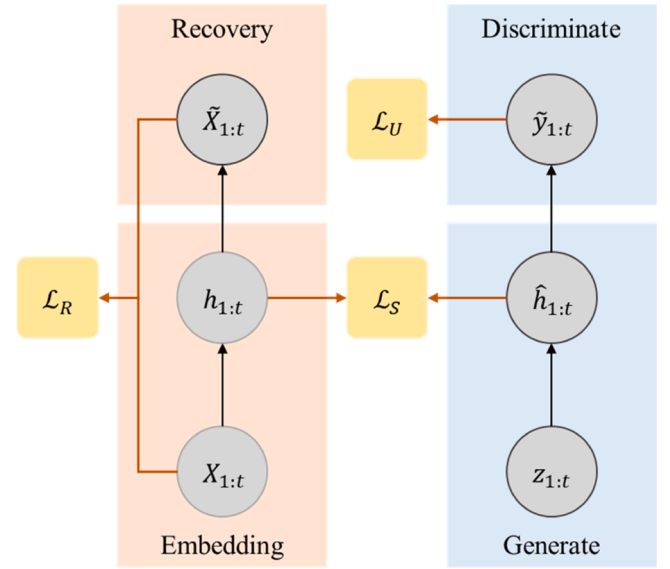


Fig. 1. TimeGAN Framework. Note: The components of the TimeGAN architecture, including its loss functions and network structure.

layer maps the data ($X_{1:t}$) into a latent space, assisting the model in learning the temporal dynamics of the data. The recovery layer restores the embedded data ($h_{1:t}$) back into the feature space ($\tilde{X}_{1:t}$). Instead of directly generating into the feature space, the generator takes a random vector ($z_{1:t}$) as input and generates into the embedding space ($\hat{h}_{1:t}$). The discriminator operates in the embedding space and classifies the input data as real or generated ($\tilde{y}_{1:t}$).

To effectively capture the characteristics of time series data, TimeGAN combines four components and introduces three loss functions. The reconstruction loss ($\mathcal{L}_R$) is utilized to guarantee the accurate encoding and decoding of the original data by the embedding layer and recovery layer. The L2-norm of the difference between the real data (x_t) and the data after passing through the embedding layer and recovery layer ($\tilde{x}_{1:t}$) is used as the loss. The unsupervised loss ($\mathcal{L}_U$) is similar to the loss used in traditional GANs. The vectors of y_t and $\tilde{y}_t$ are the discriminator's predictions for the real data and the generated data, respectively. If the discriminator predicts that the data are real, it will be close to 1. The supervised loss ($\mathcal{L}_S$) is designed to aid the generator in learning temporal relationships from the data by comparing the generated data at each time step to the actual data. The generator ($g_x(h_{t-1}, z_t)$) takes the embedded data (h_{t-1}) from the previous time point and a new random vector (z_t) to generate a value, which is then compared with the true embedded data (h_t). The loss functions used by TimeGAN are defined in Table 1.

The learning process of the embedding network (θ_e) and the recovery network (θ_r) involves the use of $\mathcal{L}_S$ and $\mathcal{L}_R$. On the other hand, the training of the generator (θ_g) and discriminator (θ_d) employs $\mathcal{L}_S$ and $\mathcal{L}_U$. This enables the acquisition of a generative model that can learn both temporal characteristics and the relationships among multiple features. Table 2 shows the optimization functions used in the learning and training processes in TimeGAN. In this paper, the loss functions—including $\mathcal{L}_S$ and $\mathcal{L}_S$ —and the networks are configured

Table 1
Loss Function for TimeGAN.

Loss Function	Formula
Reconstruction Loss	$\mathcal{L}_R = E_{X_{1:T} \sim p} [\sum_t \ x_t - \tilde{x}_t\ _2]$
Unsupervised Loss	$\mathcal{L}_U = E_{X_{1:T} \sim p} [\sum_t \log(y_t)] + E_{X_{1:T} \sim \hat{p}} [\sum_t \log(1 - \tilde{y}_t)]$
Supervised Loss	$\mathcal{L}_S = E_{X_{1:T} \sim p} [\sum_t \ h_t - g_x(h_{t-1}, z_t)\ _2]$

Table 2
Optimization Function for TimeGAN.

Networks	Optimization Function
Embedding, Recovery	$\min_{\theta_e, \theta_r} (\lambda \mathcal{L}_S + \mathcal{L}_R)$
Generator, Discriminator	$\min_{\theta_g} (\eta \mathcal{L}_S + \max_{\theta_d} \mathcal{L}_D)$

according to the settings described above. In Section 5, we will compare the proposed model with existing time-series GAN-based generation models, such as RNN-GAN and LSTM-GAN.

3.2. 3D-CNNpred model

Hoseinzade & Haratizadeh [44] proposed a convolutional neural network (CNN)-based framework for stock market prediction, as shown in Fig. 2. This framework encompasses the extraction of features from diverse sources, including various markets. Among the models they introduced, the 3D-CNNpred model utilizes a three-dimensional tensor to incorporate information from multiple markets simultaneously. The dataset comprises data from i distinct markets, k variables, and information from the past j days.

The 3D-CNNpred model proceeds with final prediction after undergoing daily feature extraction and durational feature extraction. In daily feature extraction, it computes 1×1 convolutional filters ($w_{1 \times 1}$) with a depth of k on the 3D input tensor ($v_{i,j,k}$), serving as a feature selection and extraction algorithm for the data. In durational feature extraction, filters of $3 \times$ (the number of markets) are applied to summarize data fluctuation patterns over time. Subsequently, utilizing several max pooling and convolutional layers, the extracted feature maps are employed for the ultimate prediction.

3.3. TG-3DCNN model

This paper introduces the TG-3DCNN model, which combines TimeGAN and 3D-CNNpred. TG-3DCNN is designed to handle time series data characterized by substantial volatility. The proposed model leverages past data to train TimeGAN to recognize the temporal patterns inherent in time series data. Using the trained TimeGAN model, synthetic data and real data are stacked in a 3D format and input into the prediction model.

The data used for TimeGAN training consist of a 2D dataset with a shape of a $(t \times k)$ matrix, encompassing information for k variables over t time steps. For the generation process, a random variable z is drawn from a uniform distribution in the range $[0,1]$ with the shape of a $(t \times k)$ matrix. The random variable z is passed through an encoder, resulting in an embedding matrix with a shape of $(t \times e)$. This mapping into the latent

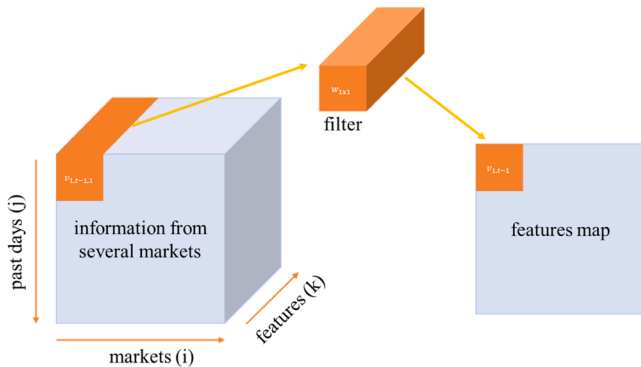


Fig. 2. Applying a Filter to a 3D Input Tensor. Note: The time-series data, originally composed of past days and features, is expanded to three dimensions by adding a markets dimension representing diverse market data. Subsequently, a CNN model is applied, where the filter dimension corresponds to the features.

space facilitates the adversarial network's ability to learn the temporal dynamics of the data through low-dimensional representations. By inputting the data into an encoder configured to a specific dimension, we can generate augmented data from the latent vector rather than from the original time-series data, which contains a lot of noise.

The overall process of the proposed model for predicting futures price movement direction is illustrated in Fig. 3. First, based on the TimeGAN model trained on the Train period data, a large number of synthetic data are generated for the Test period data. These synthetic data are then structured in a 2D form, such as $(t \times k)$, and A-1 of them are selected. Here, A-1 (augmentation-1) synthetic data are chosen based on their similarity to real data, starting with the data most similar to the original data and stacking them sequentially in the form of a 3D tensor. The 3D tensor has the form of $(t \times k \times A)$, and the mean squared error (MSE) is used as the similarity metric to ensure that augmented data similar to the original data are used. Finally, this 3D tensor is used as input for the 3DCNN to predict whether the price will rise or fall the next day.

The 3D stacking method is beneficial for the following reasons: 1) Augmenting data to increase sample size typically treats augmented data independently, meaning that the interactions between similar sequences cannot be effectively communicated to the model. 2) The 3D stacking approach enables the model to process interactions between similar sequences in a dependent manner by utilizing augmented data generated to resemble the original data. Instead of relying on a single randomly determined value in noisy time-series data, the model analyzes in a dependent manner within a dimension composed of a combination of randomly selected values most similar to the original. This allows the model to preserve the inherent temporal correlations of the data while simultaneously learning to be robust against noise.

The prediction model utilizes a 3DCNN architecture, which is highly suitable for multidimensional time-series analysis as it effectively captures spatial and temporal features along with their correlations. By discerning the correlations between real and synthetic data, the model better captures the temporal characteristics and patterns inherent in time-series data compared to using only real data as input.

4. Empirical study

The experimental process proceeds in the same order as in Fig. 4. First, we define the input data for the KOSPI200, S&P500, NASDAQ100 future and perform time series data augmentation using TimeGAN. The augmented data are expanded in dimensions to fit our proposed model, the 3DCNN. Additionally, augmented data samples are compiled for the baseline model. Subsequently, we search for the optimal hyperparameters to achieve the best learning outcome and conduct an investment simulation of the model over a period of 5 years using a sliding window approach.

4.1. Step 1: input data & evaluation metrics

The data used in this study consist of price, trading volume, and technical indicators for KOSPI200, S&P500, NASDAQ100 future, encompassing a total of 37 variables. KOSPI200 futures are futures trading on an index that is a weighted average of the top 200 companies listed on the Korean Exchange based on market capitalization. The sample period of the dataset is from July 13, 1996, to August 31, 2022. Similarly, S&P 500 and NASDAQ-100 futures are based on the market capitalization-weighted averages of the 500 largest publicly traded companies in the U.S. and the 100 largest non-financial companies listed on the NASDAQ exchange, respectively. The sample periods for each dataset are from September 10, 1997, to August 31, 2022, and from June 22, 1999, to August 31, 2022, respectively.

To ensure consistent data processing, all the variables undergo normalization through oscillator transformation. This transformation is implemented to effectively address potential distribution shift issues

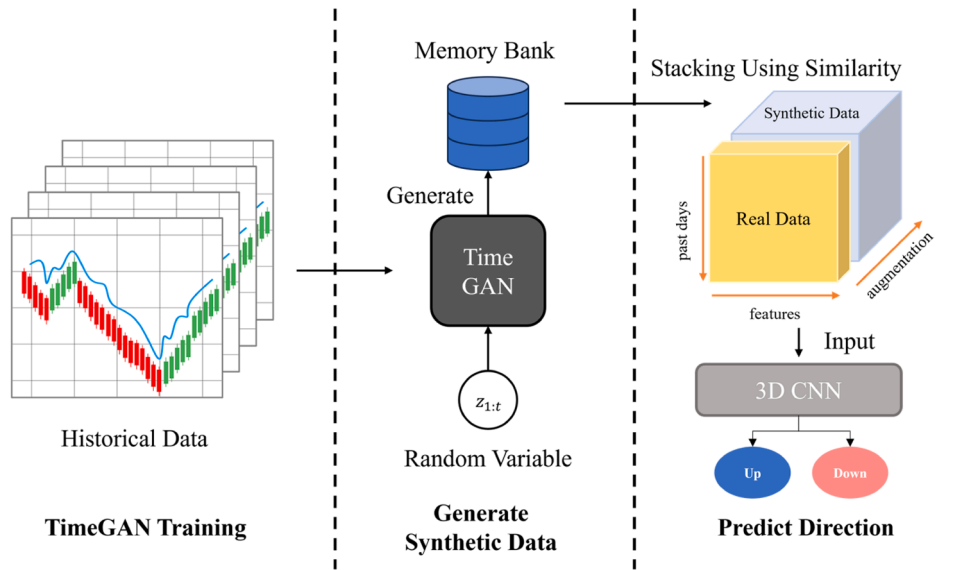


Fig. 3. Overall Process of TG-3DCNN. Note: After training the TimeGAN model on historical data, a vast number of generated data samples are stored in a memory bank. Subsequently, the generated data most similar to the data from the prediction period are sequentially stacked along the augmentation dimension to create three-dimensional data. This 3D data is then input into a 3D CNN model to perform the final task of predicting stock price movements.

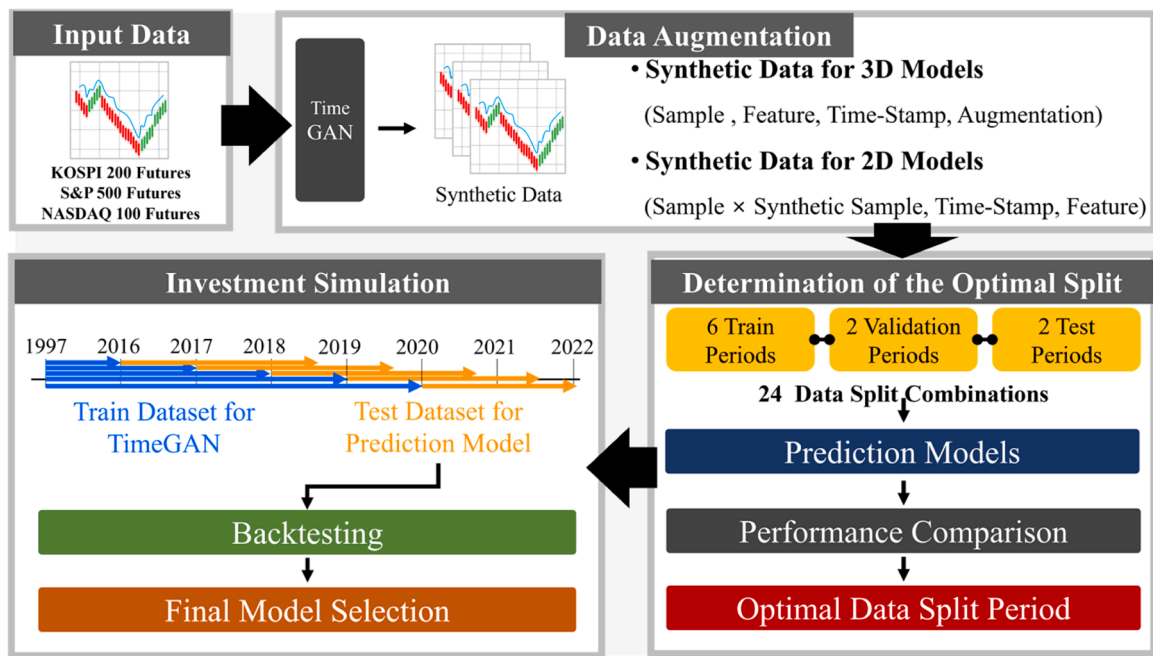


Fig. 4. Overall Experimental Process. Note: The input data is augmented according to each method, followed by hyperparameter tuning, to conduct the final investment simulation.

that might arise when transitioning from training data to test data. Detailed information about the variables employed in this study can be found in Table 3.

To systematically evaluate the performance of the model, we employ a total of five key evaluation metrics: accuracy, returns, cumulative returns, volatility, and the Sharpe ratio.

Accuracy, which is used to assess the model's ability to predict stock price movements correctly, is measured by the hit rate. The hit rate is defined as follows.

$$\text{Hit Rate} = \frac{\text{True Positive} + \text{True Negative}}{\text{True Positive} + \text{True Negative} + \text{False Positive} + \text{False Negative}} \quad (1)$$

Subsequently, to evaluate the backtest results in the actual market, we use evaluation metrics focusing on profit and risk. Returns serve as a pivotal indicator for measuring the profitability of the investment strategy, while volatility is employed to gauge the level of risk by indicating the extent of price fluctuations.

The Sharpe ratio, an essential metric, is used to provide a balance between investment returns and risk, enabling an assessment of the efficiency of the investment strategy. In this context, the interest rate of

Table 3
Description of Variables.

#	Variable	Description
1	MACD	Subtracts the long-term exponential moving average from the short-term exponential moving average (Short: 12, Long: 26)
2	Price ROC (%)	Difference between current stock price and stock price n period ago (Short: 10, Long: 20)
3	Sonar	Marginal rate of change of moving average line (25 days)
4	Stochastic RSI	Indicator of the current relative position of RSI within the RSI fluctuation range over a certain period of time (RSI: 14, H-L: 7)
5	Tipple Exponential Smoothed Moving Average	Growth rate of the value obtained by performing the exponential moving average 3 times (12 days)
6	Chaikin Volatility	Indicator of the rate of change in price fluctuations (MA: 10, Ratio: 10)
7	Stochastics Oscillator	Indicator showing the current price position as a percentage within the range of the highest and lowest prices over the last N days (SlowK: 5, SlowD: 5, FastK: 12)
8	Chaikin Oscillator	Oscillator that subtracts the long-term ADI moving average from the short-term ADI moving average (Short: 3, Long: 10)
9	Detrended Price Oscillator	Eliminates long-term price trends by using moving averages in prices (14 days)
10	Ease of Movement	Indicator of the correlation between stock price and trading volume (10 days)
11	Mass Index	Fluctuation between high and low prices (Term: 9, Signal: 25)
12	Net Change Oscillator	Indicator used to confirm short-term trends using a market analysis method using momentum (12 days)
13	Price Oscillator	Difference between two exponential moving averages of different periods (Short: 10, Long: 20)
14	Relative Strength Index	Indicator of relative strength between upward and downward price pressures by comparing the average upward and downward amplitude of stock prices (14 days)
15	Stochastics Fast	Indicator that captures short-term volatility in prices well by calculating short-term moving averages on stock charts (5 days)
16	Stochastics	Indicator of the relative strength of stock price movements (12 days)
17	William 'R	Moves between 0 and -100 and measures overbought and oversold levels (14 days)
18	Money Flow Index	Measures the relative strength of buying and selling (14 days)
19	Chaikin Money Flow	Uses both the intensity of buying and selling pressure and trading volume (20 days)
20	Volume ROC	Difference between today's trading volume and trading volume n days ago (12 days)
21	Stochastics Slow	Indicator supplemented by increasing the sensitivity and stability of Stochastic fast (5 days)
22	Disparity 20	Indicator of the gap between the stock price and the moving average line (On a 20-day basis)
23	Disparity 60	Indicator of the gap between the stock price and the moving average line (On a 60-day basis)
24	Zscore (Sigma)	Indicator of how many standard deviations a given data point is from the mean (20 days)
25	Sonar Signal	Time series chart developed to identify turning points in the stock cycle (9 days)
26	Commodity Channel Index	Auxiliary indicator that measures the difference between stock price and moving average (9 days)
27	Sonar2	Indicator to detect trend reversal in stocks (25 days)
28	Moving Average Oscillator	Moving average oscillator indicator created based on Accumulation/Distribution indicator (Short: 5, Long: 20)
29	S-ROC	Indicator using smoothed exponential moving average (Period: 10, N: 20)
30	Force Index Short	Indicator of upward price pressure in the market (stock price movement over a short period of time; 2 days)
31	Force Index Long	Indicator of upward price pressure in the market (stock price movement over a long period of time; 13 days)
32	Psychology	Indicator of overheating and recession situations through changes in investment sentiment (10 days)
33	Volume Oscillator	Trading volume: Separation between the short-term moving average and the long-term moving average (Long: 20, Short: 10)
34	Reverse	Momentum indicator that moves in the opposite direction from the stock price (12 days)
35	Relative Momentum Index	Indicator that analyzes the point of trend change in the form of a development in the RSI indicator (M: 14, N: 5)
36	Band %B	Indicator quantified by comparing the upper and lower limits of Bollinger Bands (DevRatio: 2, Term: 20)
37	MACD Oscillator	Indicator that finds the moment of intersection during the repetition of convergence and divergence of moving lines (Short: 12, Long: 26)

Korean government bonds of 2.5 % is employed as the annual risk-free rate.

Furthermore, cumulative returns, which indicate the cumulative profits and losses from the inception of the investment to the present relative to the initial investment amount, play a critical role in evaluating the strategy's overall performance. Cumulative returns were continually updated by multiplying the daily returns with the existing cumulative returns, allowing for a comprehensive assessment of the investment strategy.

We evaluate the model performance and backtesting results through these various metrics. The corresponding formulas are listed in Eqs. (1–5).

$$\text{Cumulative Returns} = \prod_{i=0}^N (1 + \text{daily returns}) - 1 \quad (5)$$

4.2. Step 2: data augmentation

Table 4 provides a comparative overview of the characteristics of the proposed model and the baseline model. The 3-dimensional extended models include our proposed TG-3DCNN (TimeGAN) model, as well as RG/LG-3DCNN (RNN-GAN and LSTM-GAN), the unaugmented model is a general time series deep learning model without GAN-based models, and the sample augmented model represents a model that utilizes augmented data as samples. In this experiment, GAN-based models

$$\text{Monthly Return} = \frac{\text{Adj close}(\text{end of month}) - \text{Adj close}(\text{begining of month})}{\text{Adjusted close at the begining of the month}} \times 100\% \quad (2)$$

$$\text{Volatility} = \sqrt{\frac{\sum_{i=1}^n (\text{daily returns} - \text{average daily returns})^2}{n - 1}} \quad (3)$$

$$\text{Sharpe Ratio} = \frac{\text{Rate of Return} - \text{Risk Free Rate}}{\text{Standard Deviation of Return}} \quad (4)$$

conduct a cumulative training process five times (5 testing years), starting from 20 years, and incrementally adding one year's worth of data each time. After the 5-year testing period, one year's worth of data is set as the investment performance evaluation period of the prediction model. Each training is set to 100,000 epochs.

The data generated through GAN-based models have an input form such as (131072, 20, 37) for the proposed model, and for the sample

Table 4
Comparison Between the Proposed Model and the Baseline Model.

Model Categorization	Model	Description
3-dimensional Extended (Our model)	10/100/500TG/RG/LG-3DCNN	Used A-1 augmentations to stack in 3-D and employed a 3DCNN model as the prediction model
Unaugmented	3DCNN, 2DCNN, LSTM	Used 3DCNN, 2DCNN, and LSTM models as prediction models without augmentation
Sample Augmented	10/100/500TG/RG/LG-2DCNN, 10/100/500TG/RG/LG-LSTM	Used A-1 augmented data as samples

augmented model, the time period is set to 21 because the y value must be generated during the augmentation process. Therefore, an input form is (131072, 21, 37) for the sample augmented model. In this form, 131072 represents the number of samples in the memory bank, 20 denotes the time period, which is the number of past data used for prediction, and 37 represents the number of variables. The similarity of the generated data is assessed based on the mean squared error (MSE) compared to the original data, and the top 9, 99, and 499 data points with the lowest loss values are selected. These selected augmented data are combined with the original sample data in an A-1 format, resulting in a final input format of sample, feature, time period, and augmentation.

On the other hand, one of the baseline models, the sample augmented model, cannot directly apply dimension augmentation. Instead, it augments the data by expanding the rows of the dataset. As a result, the input data format differs from that of the 3DCNN, resulting in a final input form of (sample $\times$ augmentation, time period, feature). The specific hyperparameter settings for the Proposed Model and Baseline Model are presented in Table A1 of appendix. This conclusion was derived by exploring various combinations of hyperparameters using a GridSearch approach on the validation data. The number of linear layers was set to 1, 2, 4, and 8, the hidden dimensions to 8, 16, 32, and 64, and the learning rate to 0.001, 0.0005, and 0.0001. For CNN-based models, the kernel and filter sizes were appropriately adjusted for each setting. To prevent overfitting, early stopping with a patience of 5 was implemented, meaning training would stop if there was no improvement in the validation loss for 5 consecutive attempts. In future experimental settings, a sliding window approach will be applied to 5 years of data, and the final model will be derived based on the average results across all periods. This approaches ensure that the model is not solely focused on minimizing loss in the training data but is also capable of demonstrating robust performance on unseen data. Additionally, given the extreme variability in investment performance across different periods, which is characteristic of financial time-series data, evaluating the model over multiple years enhances the reliability of the results.

4.3. Step 3&4: experimental setting

GAN-based models commence its training with a 20-year dataset, spanning from January 8, 1997, to December 31, 2016, and subsequently rebalances and accumulates data on an annual basis. Therefore, GAN-based models learning for evaluation have a total of 5 testing years, from models learned from 2016 to 2020. The performance evaluation assesses the model's ability to predict the upward or downward movement of the next day's futures index for each respective period.

Determination of the Optimal Split. To precisely measure the performance of the prediction models utilizing augmented time series data generated by TimeGAN, we conduct cross-validation using the sliding window technique. Prior to testing the prediction models, we determine the optimal training-validation-test period to achieve the best model performance. We consider 24 combinations, where the training period ranges from 3, 6, 9, 12, 18, and 24 months; the validation period

is set to 1 or 2 months; and the test period is 1 or 2 months. To select the optimal period, we randomly sample 2 months from the 5 testing years, comparing the prediction accuracy for a total of 10 months. We determine the data split period based on the performance for these 10 months, selecting the 6:2:1 ratio, which exhibits the best prediction performance among the aggregated data for different ratios. The details of the randomly selected periods are provided in Table 5 below.

Investment Simulation. Following the determination of the data split period, we select the optimal algorithmic trading model for futures investment. In the stock market, actual assets are traded, while in the futures market, contracts about future prices are traded. Margin is a deposit required when a trader trades stocks or futures, which is a certain percentage of the contract amount. Especially in the futures market, due to the potential for large losses from price fluctuations, there is a system called 'margin call' that requires additional margin depending on price changes. To do so, we backtest prediction models based on previously described performance metrics, including returns, volatility, and the Sharpe ratio. Notably, our experiments incorporate the mechanism of margin calls, which adjust according to price fluctuations. Backtesting is performed using each model's signals in a real futures market environment, and the monthly investment performance, 5-year cumulative returns, costs, and other factors are used to select the final model. Detailed information on the period used in the experiment is shown in Table 6 below. Similar to the data split period phase, TimeGAN is initially trained on 20 years of data for the testing year and subsequently rebalances and accumulates data on an annual basis. The predictive models are evaluated in a sliding window by using a 6:2:1 period for one year after the TimeGAN learning period. Therefore, investment performance over 5 years is evaluated for a total of 5 testing years, one year for each testing year. The selected 5 testing years were chosen because the remaining period, after excluding the 20 years required for sufficient training of TimeGAN, amounts to 5 years. This 5-year period is sufficiently long to yield meaningful results; however, it has the limitation of not fully accounting for the diverse market conditions that have occurred historically (e.g., the 2008 financial crisis). Nevertheless, since the 5 testing years include the COVID-19 pandemic period, which encompasses both recessionary and booming phases, this limitation is partially mitigated.

5. Experimental results

5.1. Determination of the optimal data split period

Table 7 displays the hit ratio results for data split periods based on the established experimental settings. The model performance is highest, approximately 65 %, for the 3:2:1 and 6:2:1 split. When the periods of 3:2:1 and 6:2:1 are arranged in descending order, the 6:2:1 period demonstrates superior results for top-rank models (those above 50 %), and thus, it is selected as the setting for the final evaluation of model performance.

Table 5
Randomly Selected Periods for Ratio Determination.

5 Testing Years	Start Date of Sample
1997–2016	2018/04
	2019/08
1997–2017	2019/03
	2019/05
1997–2018	2020/11
	2019/04
1997–2019	2020/04
	2020/12
1997–2020	2021/01
	2021/07

Table 6
Investment Simulation Setting Based on 5-Year Futures Data.

Year	TimeGAN Training	Training (6 months)	Validation (2 months)	Test (1 month)
2016	1997/01/08–2016/12/31 (20 Years)	2017/01/01–2017/06/30	2017/07/01–2017/08/30	2017/09/01–2017/09/30
		2017/02/01–2017/07/31	2017/08/01–2017/09/30	2017/10/01–2017/10/31
		...		
2017	1997/01/08–2017/12/31 (21 Years)	2017/11/01–2018/04/31	2018/05/01–2018/06/30	2018/07/01–2018/07/31
		2017/12/01–2018/05/31	2018/06/01–2018/07/31	2018/08/01–2018/8/31
		2018/01/01–2018/06/30	2018/07/01–2018/08/31	2018/09/01–2018/09/30
2018	1997/01/08–2018/12/31 (22 Years)	2018/02/01–2018/07/31	2018/08/01–2018/09/30	2018/10/01–2018/10/31
		...		
		2018/11/01–2019/04/31	2019/05/01–2019/06/30	2019/07/01–2019/07/31
2019	1997/01/08–2019/12/31 (23 Years)	2018/12/01–2019/05/31	2019/06/01–2019/07/31	2019/08/01–2019/8/31
		2019/01/01–2019/06/30	2019/07/01–2019/08/31	2019/09/01–2019/09/30
		2019/02/01–2019/07/31	2019/08/01–2019/09/30	2019/10/01–2019/10/31
2020	1997/01/08–2020/12/31 (24 Years)	2019/11/01–2020/04/31	2020/05/01–2020/06/30	2020/07/01–2020/07/31
		2019/12/01–2020/05/31	2020/06/01–2020/07/31	2020/08/01–2020/8/31
		2020/01/01–2020/06/30	2020/07/01–2020/08/31	2020/09/01–2020/09/30
		2020/02/01–2020/07/31	2020/08/01–2020/09/30	2020/10/01–2020/10/31
		...		
		2020/11/01–2021/04/31	2021/05/01–2021/06/30	2021/07/01–2021/07/31
		2020/12/01–2021/05/31	2021/06/01–2021/07/31	2021/08/01–2021/8/31
		2021/01/01–2021/06/30	2021/07/01–2021/08/31	2021/09/01–2021/09/30
		2021/02/01–2021/07/31	2021/08/01–2021/09/30	2021/10/01–2021/10/31
		...		
		2021/11/01–2022/04/31	2022/05/01–2022/06/30	2022/07/01–2022/07/31
		2021/12/01–2022/05/31	2022/06/01–2022/07/31	2022/08/01–2022/8/31

5.2. Algorithmic trading performance

Based on the optimal hyperparameters determined in the previous section (GAN-based models trained on 20 years of data with rebalancing

and cumulative training every year, and the prediction model trained with a 6:2:1 period, i.e., rebalancing every month), the final performance evaluation is conducted. Table 8 displays the results based on the overall average performance of KOSPI 200, S&P 500, and NASDAQ 100

Table 7
Comparing the Model Performance Based on the Data Split Ratio.

Model	311ACC	312ACC	321ACC	322ACC	611ACC	612ACC	621ACC	622ACC
3DCNN	0.581	0.603	0.610	0.576	0.556	0.565	0.626	0.587
10TG–3DCNN	0.617	0.588	0.579	0.557	0.546	0.582	0.582	0.550
100TG–3DCNN	0.559	0.572	0.553	0.554	0.555	0.575	0.610	0.564
500TG–3DCNN	0.601	0.603	0.594	0.569	0.554	0.576	0.641	0.584
2DCNN	0.581	0.603	0.610	0.576	0.556	0.565	0.626	0.587
10TG–2DCNN	0.524	0.511	0.485	0.467	0.461	0.442	0.441	0.458
100TG–2DCNN	0.539	0.513	0.506	0.480	0.498	0.464	0.430	0.462
500TG–2DCNN	0.497	0.473	0.485	0.472	0.439	0.464	0.473	0.512
LSTM	0.610	0.625	0.650	0.609	0.603	0.606	0.553	0.555
10TG-LSTM	0.446	0.493	0.469	0.454	0.579	0.478	0.365	0.410
100TG-LSTM	0.523	0.511	0.437	0.447	0.484	0.455	0.404	0.427
500TG-LSTM	0.526	0.515	0.489	0.504	0.520	0.496	0.488	0.483
Model	911ACC	912ACC	921ACC	922ACC	1211ACC	1212ACC	1221ACC	1222ACC
3DCNN	0.462	0.505	0.527	0.485	0.530	0.487	0.468	0.474
10TG–3DCNN	0.464	0.491	0.462	0.471	0.505	0.509	0.537	0.510
100TG–3DCNN	0.469	0.508	0.466	0.466	0.491	0.511	0.464	0.484
500TG–3DCNN	0.434	0.491	0.520	0.491	0.453	0.466	0.519	0.508
2DCNN	0.462	0.505	0.527	0.485	0.530	0.487	0.468	0.474
10TG–2DCNN	0.544	0.486	0.436	0.500	0.511	0.501	0.454	0.497
100TG–2DCNN	0.477	0.480	0.450	0.489	0.541	0.532	0.531	0.554
500TG–2DCNN	0.420	0.447	0.513	0.501	0.529	0.486	0.478	0.513
LSTM	0.425	0.461	0.567	0.513	0.511	0.491	0.475	0.472
10TG-LSTM	0.425	0.438	0.453	0.511	0.474	0.427	0.387	0.480
100TG-LSTM	0.432	0.468	0.520	0.518	0.567	0.499	0.425	0.486
500TG-LSTM	0.393	0.416	0.420	0.510	0.568	0.498	0.413	0.488
Model	1811ACC	1812ACC	1821ACC	1822ACC	2411ACC	2412ACC	2421ACC	2422ACC
3DCNN	0.546	0.528	0.490	0.512	0.581	0.553	0.531	0.540
10TG–3DCNN	0.532	0.521	0.557	0.533	0.434	0.454	0.510	0.529
100TG–3DCNN	0.529	0.523	0.498	0.540	0.496	0.529	0.576	0.585
500TG–3DCNN	0.552	0.496	0.467	0.485	0.455	0.489	0.557	0.576
2DCNN	0.546	0.528	0.490	0.512	0.581	0.553	0.531	0.540
10TG–2DCNN	0.517	0.493	0.459	0.479	0.443	0.474	0.504	0.452
100TG–2DCNN	0.505	0.504	0.512	0.514	0.444	0.421	0.405	0.437
500TG–2DCNN	0.507	0.500	0.462	0.482	0.500	0.459	0.420	0.433
LSTM	0.524	0.497	0.475	0.489	0.516	0.556	0.576	0.572
10TG-LSTM	0.506	0.464	0.428	0.473	0.468	0.485	0.550	0.497
100TG-LSTM	0.487	0.456	0.455	0.497	0.522	0.490	0.461	0.462
500TG-LSTM	0.481	0.504	0.506	0.491	0.537	0.489	0.399	0.437

Note: The highlighted sections represent the best performance in each period.

Table 8
5-Year Monthly Performance Comparison of Algorithmic Trading in average of all market.

Model	Volatility (%)	Returns (%)	Sharpe Ratio	Cumulative Returns (%)	Hit Ratio (Accuracy)	Learning Time (s)	Data Size (KB)
10TG-3DCNN	3.223	1.644	0.769	2124.445	0.560	1.786	248941
3DCNN	3.293	<u>1.557</u>	<u>0.568</u>	2537.145	0.554	0.976	4188
2DCNN	3.293	<u>1.557</u>	<u>0.568</u>	2537.145	0.539	1.017	4188
LSTM	3.170	1.513	0.545	1643.228	0.530	2.132	4188
100TG-LSTM	2.939	0.972	0.467	1472.445	0.542	6992.823	2683899
100TG-3DCNN	3.265	0.988	0.464	1189.048	0.528	19.438	2489407
500TG-LSTM*	2.936	0.797	0.444	1123.458	0.535	44656.755	13419493
10RG-3DCNN	3.972	1.138	0.426	1747.932	0.535	1.786	248941
10TG-2DCNN* **	2.794	0.430	0.408	-573.745	0.551	233.584	268390
100LG-2DCNN	3.807	1.229	0.397	1784.915	0.534	2442.125	2683899
500TG-2DCNN* **	2.827	0.419	0.366	-333.225	0.555	11414.911	13419493
100LG-3DCNN	4.116	1.248	0.346	2032.425	0.537	19.438	2489407
100TG-2DCNN*	3.442	0.835	0.326	155.270	<u>0.557</u>	2442.125	2683899
10TG-LSTM*	<u>2.805</u>	0.622	0.320	1016.572	0.514	582.858	268390
10LG-3DCNN	3.865	1.076	0.318	1499.332	0.547	1.786	248941
500TG-3DCNN	3.287	1.003	0.312	839.975	0.543	165.441	12477032
500LG-LSTM	3.596	1.011	0.298	1238.022	0.521	44656.755	13419493
10LG-2DCNN* *	3.810	0.520	0.290	-922.435	0.535	233.584	268390
10RG-2DCNN	3.714	0.748	0.191	650.432	0.529	233.584	268390
100LG-LSTM	3.685	0.892	0.106	1715.122	0.528	6992.823	2683899
10RG-LSTM	3.788	0.826	0.004	994.975	0.526	582.858	268390
10LG-LSTM* **	3.582	0.794	-0.029	-71.032	0.521	582.858	268390
100RG-2DCNN	3.786	1.204	-0.039	1425.548	0.538	2442.125	2683899
500LG-2DCNN	4.253	0.967	-0.316	806.525	0.525	11414.911	13419493
100RG-LSTM	3.610	0.802	-0.319	974.822	0.519	6992.823	2683899
500RG-LSTM* *	3.485	0.596	-0.335	379.522	0.524	44656.755	13419493
100RG-3DCNN	4.011	1.047	-0.621	1512.265	0.528	19.438	2489407
500RG-2DCNN	3.974	1.157	-0.686	1533.762	0.525	11414.911	13419493
500RG-3DCNN	4.173	1.411	-0.849	<u>2130.695</u>	0.524	165.441	12477032
500LG-3DCNN	3.646	1.030	-1.035	750.508	0.545	165.441	12477032
Buy & Hold* **	0.738	0.141	-1.464	15.522	0.532	-	-

Note: The bolded sections represent the best performance across each metric, and the underlined sections indicate the second-best performance across each metric. To evaluate the statistical significance of the performance results between the proposed model and comparison models, pairwise t-tests were conducted for each market and evaluation metric. Since the sample size was 20 and normality was not satisfied, the Mann-Whitney *U* test was performed instead. The symbols * denote significance at the 10 % level, * * at the 5 % level, and * ** at the 1 % level, while the absence of these symbols indicates that the results were not statistically significant.

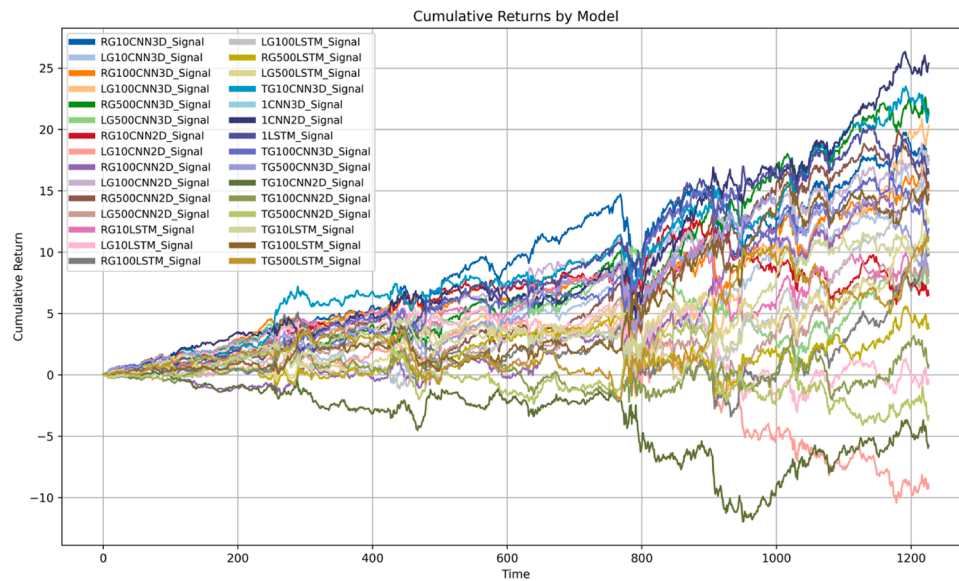


Fig. 5. 5-Year Cumulative Returns by Model in average of all markets. Note: A graph of the average cumulative returns during the investment simulation period for each method.

futures, including the monthly average investment performance over a period of 5 years, cumulative return rates for the 5-year period, training time, and storage size for the trained data of each model. The results for each market can be found in Table A of the Appendix. The table is sorted in descending order based on the Sharpe ratio. Interpreting the overall results, the models are ranked as follows: 3-dimensional extended,

unaugmented, and sample augmented. The performance of GAN-based models improves significantly as the number of augmentations decreases. This can be interpreted as artificial data becoming excessively biased as it increases in higher dimensions or larger samples. Among GAN-based models, TimeGAN demonstrates superior performance, whereas RNN-GAN and LSTM-GAN show relatively weaker results. This

is attributed to the unique characteristics of TimeGAN, such as augmenting embedded data and employing supervised loss to generate sequences closely resembling actual data. Synthesizing these findings, it can be concluded that better performance is achieved with fewer augmentation iterations, using TimeGAN as the augmentation model, and applying a 3-dimensional extended approach. Notably, the model with the highest performance, 10TG-3DCNN, provides consistent and clear results, reinforcing the validity of these observations.

While most models underperform compared to the unaugmented models, the proposed 10TG-3DCNN model performs significantly better. This model records a monthly average return of 1.644 %, which translates to an annual return of approximately 19.728 %. The model's return volatility is around 3.223 %, achieving an excess return of 0.769 times relative to the risk, which is approximately 1.35 times higher than the second-best performance of 0.568. Additionally, when examining the learning time and data size, the proposed model, which augments by expanding into new dimensions, proves to be more cost-efficient in terms of storage compared to the sample-augmented models. The learning process is also cost-efficient as it captures the multidimensional spatial structure through expanded dimensions. This is evident from the learning time metric, which shows that the model can train up to 6390 times faster, in approximately 1.8 seconds.

Fig. 5 displays the results of the 5-year cumulative return based on the overall market average for each model. The proposed model consistently outperforms the others in most periods. When compared with the cumulative returns shown in Table 8, the proposed model (in turquoise) consistently delivers superior performance throughout most periods. When evaluating the final 5-year cumulative return, it exhibits an impressive performance of approximately 2124.445 % compared to the baseline. Although it does not achieve the highest cumulative return, the difference from the second-highest return of 2130.695 % is negligible, and the graph shows that in the early stages, it outperforms significantly. Therefore, our proposed model excels in not only short-term investments but also long-term strategies. From the perspectives of storage capacity and training cost, it can be considered an extremely efficient optimal algorithmic trading system.

6. Conclusion

In this paper, we propose a new algorithmic trading model targeting the KOSPI200 futures market. By applying the time series augmentation technique TimeGAN and extending it into 3 dimensions using 3DCNN, we develop a robust, superior, and efficient model that effectively captures multidimensional information from the data. According to the experimental results, the proposed 3-dimensional extended model outperforms the traditional unaugmented stock prediction model. Additionally, the sample augmentation model, which increases the data sample count through time-series augmentation, fails to surpass the baseline model, and existing time-series augmentation methods such as RNN-GAN and LSTM-GAN demonstrate lower performance compared to TimeGAN, as they do not effectively capture the complexity of time-series data. The proposed 10TG-3DCNN model achieves the highest short-term returns and risk-adjusted returns, while also securing significant long-term gains. This is evidenced by a performance improvement of approximately 1.35 times compared to the second-best model in terms of risk-adjusted returns. Additionally, it demonstrates up to 6390 times greater efficiency in terms of data size and training time, making it more efficient than other models and a practical solution for both performance improvement and cost reduction.

Financial markets feature various traded financial instruments, with futures transactions being particularly appealing to investors in these markets. This paper demonstrates that simple data augmentation can lead to reduced performance and increased costs. In contrast, expanding the augmentation into three dimensions shows greater cost-effectiveness

and better performance compared to existing methods. This suggests that using the proposed model to predict futures market movements, based on TimeGAN and 3D-CNN, allows investors in various futures markets to create more efficient portfolios. Additionally, we perform backtesting, reflecting real futures market conditions such as margins and contract sizes, to evaluate its applicability to index futures, which are beneficial for both investing and hedging. This approach aims to provide valuable investment guidance to investors involved in futures trading.

One limitation of this study is that, due to the computational demands of augmentation, we were unable to explore various augmentation sizes extensively. From the perspective of dimensional expansion, experiments with augmentation sizes of 10, 100, and 500 revealed that models using augmentations of 100 and 500 exhibited poorer performance. This suggests that excessive augmentation may introduce substantial bias, as the augmented data is artificially generated. Consequently, future research will focus on expanding experiments with smaller augmentation sizes, using 10 augmentations as the baseline.

Additionally, training TimeGAN requires a substantial amount of data. Experiments conducted over different periods (5, 10, 15, and 20 years) showed that using 20 years of data for augmentation yielded the most robust results. This outcome is likely because the model identifies the generated data most similar to the testing period data from a large pool of candidates in the memory bank, necessitating the learning of long-term temporal patterns. Therefore, future studies should aim to develop methods that minimize the amount of data required for training augmentation models while maintaining high performance.

Moreover, future research could explore advanced predictive models that incorporate transformer-based methodologies, which are increasingly prominent in the AI field, in addition to CNN and LSTM architectures. The proposed model also holds potential for application in various financial markets, including foreign exchange, stocks, options, and real estate, offering avenues for further investigation.

CRediT authorship contribution statement

Sanghyuk Yoo: Visualization. **Kyong Joo Oh:** Software, Resources, Funding acquisition. **Heesoo Lee:** Writing – review & editing, Supervision. **Sanghoe Kim:** Writing – original draft, Methodology, Formal analysis. **Minwoo Jang:** Writing – original draft, Investigation. **Woojung Kim:** Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Jiyoung Jeon:** Writing – original draft, Methodology, Investigation.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT in order to improve language and readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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A. Appendix.

Table A1
Hyperparameter Setting for Proposed Model and Baseline Models

Method	Hyperparameters	Other Hyperparameters
10TG–3DCNN	Convolution layer 1 filter size: (8 × 1 × 1) Convolution layer 2 filter size: (16 × 3 × 10) Max pooling layer: (2 × 1) number of Linear layer: 1, hidden dimension; 72	He Initialization Adam Optimizer Learning rate: 0.001 Early Stopping patience: 5
100TG–3DCNN	Convolution layer 1 filter size: (8 × 1 × 1) Convolution layer 2 filter size: (16 × 3 × 3) Max pooling layer 1: (1 × 2) Convolution layer 3 filter size: (32 × 3 × 3) Max pooling layer 2: (1 × 2) number of Linear layer: 1, hidden dimension; 1280	
500TG–3DCNN	Convolution layer 1 filter size: (8 × 1 × 1) Convolution layer 2 filter size: (16 × 3 × 21) Max pooling layer 1: (1 × 2) Convolution layer 3 filter size: (32 × 3 × 21) Max pooling layer 2: (1 × 2) Convolution layer 3 filter size: (32 × 3 × 21) number of Linear layer: 1, hidden dimension; 2400	
2D-CNNs	Convolution layer 1 filter size: (8 × 1) Convolution layer 2 filter size: (8 × 3) Max pooling layer: (2) number of Linear layer: 1, hidden dimension; 72	
LSTMs	number of layer3, hidden dimension; 16	
TimeGAN	number of layer: 3, hidden dimension; 24, Train steps: 100,000, λ :0.1, η :100, gamma: 1	
RNN-GAN	number of layer:3, hidden dimension; 24, gamma: 1	
LSTM-GAN		

Table A2
5-Year Monthly Performance Comparison of Algorithmic Trading in Kospi 200, S&P 500, NASDAQ 100 and average of all markets

Model	KOSPI 200 futures					S&P 500 futures				
	Volatility (%)	Returns (%)	Sharpe Ratio	Cumulative Returns (%)	Hit Ratio (Accuracy)	Volatility (%)	Returns (%)	Sharpe Ratio	Cumulative Returns (%)	Hit Ratio (Accuracy)
3DCNN	1.703	1.188		72.975	0.538	3.460	1.625		2261.240	0.578
2DCNN	1.703	1.188	0.428	72.975	0.538	3.460	1.625	<u>0.640</u>	2261.240	0.548
LSTM	<u>1.672</u>	<u>1.630</u>	<u>0.973</u>	99.325	0.544	3.152	1.366	0.118	1318.500	0.525
10RG–3DCNN	3.726	0.639	0.138	78.515	0.516	3.577	1.074	0.564	658.010	0.576
100RG–3DCNN	3.331	0.698	–2.704	95.075	0.510	3.700	0.902	0.293	561.190	0.544
500RG–3DCNN	3.520	0.954	–2.437	105.335	0.521	3.875	<u>1.405</u>	–0.138	1339.500	0.501
10RG–2DCNN	3.745	–0.026	–0.146	–27.725	0.502	3.261	0.916	0.251	554.860	0.559
100RG–2DCNN	3.838	0.881	–0.540	112.975	0.510	3.310	1.395	0.557	1313.570	0.565
500RG–2DCNN	3.583	0.636	–2.514	86.625	0.521	3.387	1.316	0.301	1417.750	0.511
10RG-LSTM	3.673	0.064	–0.762	–70.575	0.504	3.330	0.991	0.506	821.850	0.517
100RG-LSTM	3.414	0.397	–1.610	6.115	0.501	3.323	0.774	0.509	459.140	0.533
500RG-LSTM	3.399	0.102	–0.757	–47.665	0.494	3.067	0.413	–0.357	–693.880	0.556
10LG–3DCNN	3.917	0.574	0.202	108.625	0.521	3.464	1.305	0.681	1083.050	<u>0.587</u>
100LG–3DCNN	3.832	0.897	0.015	146.725	0.481	3.638	1.378	0.461	<u>1563.810</u>	0.588
500LG–3DCNN	3.833	0.534	–0.012	83.025	0.513	3.731	1.387	–0.518	1464.000	0.563
10LG–2DCNN	3.661	0.347	0.114	46.375	0.504	3.399	0.665	0.362	74.980	0.569
100LG–2DCNN	3.413	0.727	0.219	79.315	0.507	3.565	0.914	0.348	620.680	0.562

(continued on next page)

Table A2 (continued)

Model	KOSPI 200 futures					S&P 500 futures				
	Volatility (%)	Returns (%)	Sharpe Ratio	Cumulative Returns (%)	Hit Ratio (Accuracy)	Volatility (%)	Returns (%)	Sharpe Ratio	Cumulative Returns (%)	Hit Ratio (Accuracy)
500LG–2DCNN	4.382	0.516		90.925	0.505	3.749	1.238		1235.000	0.535
10LG-LSTM	3.435	0.673	0.123	89.175	0.522	3.327	0.552	−0.593	−145.750	0.509
100LG-LSTM	3.490	0.460	−0.377	29.825	0.487	3.239	0.774	0.333	725.620	0.566
500LG-LSTM	3.288	0.966	0.080	116.025	0.524	3.212	0.697	0.233	162.520	0.530
10TG–3DCNN	1.599	2.235	0.112	<u>135.575</u>	0.529	3.331	0.958	0.473	833.580	0.575
100TG–3DCNN	1.978	0.146	1.297	8.625	0.535	3.563	1.072	0.451	942.950	0.524
500TG–3DCNN	1.895	0.382	0.521	24.425	<u>0.540</u>	3.526	1.348	0.560	1036.600	0.537
10TG–2DCNN	1.718	0.384	0.289	24.525	0.513	2.874	0.282	0.406	−844.550	0.564
100TG–2DCNN	1.913	−1.279	0.696	−75.025	0.498	<u>2.915</u>	0.630	0.356	18.610	0.560
500TG–2DCNN	1.733	0.143	−0.654	11.125	0.515	3.002	0.325	0.360	−367.580	0.538
10TG-LSTM	1.787	−0.043	0.419	−0.875	0.506	2.947	0.652	0.336	199.260	0.515
100TG-LSTM	1.802	0.395	0.421	27.325	0.512	3.124	1.065	0.275	661.840	0.531
500TG-LSTM	1.688	0.529	0.405	35.525	0.508	3.147	0.384	0.456	−474.100	0.499
Buy & Hold	0.738	0.141	0.403	15.522	0.532	0.738	0.141	0.348	15.522	0.532
			−1.464					−1.464		
Model	NASDAQ 100 futures					Average				
	Volatility (%)	Returns (%)	Sharpe Ratio	Cumulative Returns (%)	Hit Ratio (Accuracy)	Volatility (%)	Returns (%)	Sharpe Ratio	Cumulative Returns (%)	Hit Ratio (Accuracy)
3DCNN	4.717	1.857	0.635	<u>5277.220</u>	0.530	3.293	<u>1.557</u>	<u>0.568</u>	2537.145	0.554
2DCNN	4.717	1.857	0.635	<u>5277.220</u>	0.530	3.293	<u>1.557</u>	<u>0.568</u>	2537.145	0.539
LSTM	4.686	1.544	0.544	3511.860	0.536	3.170	1.513	0.545	1643.228	0.530
10RG–3DCNN	4.612	1.700	0.576	4507.270	0.512	3.972	1.138	0.426	1747.932	0.535
100RG–3DCNN	5.003	1.542	0.548	3880.530	0.530	4.011	1.047	−0.621	1512.265	0.528
500RG–3DCNN	5.124	<u>1.875</u>	0.028	4947.250	0.551	4.173	1.411	−0.849	<u>2130.695</u>	0.524
10RG–2DCNN	4.136	1.353	0.469	1424.160	0.525	3.714	0.748	0.191	650.432	0.529
100RG–2DCNN	4.210	1.336	−0.133	2850.100	0.538	3.786	1.204	−0.039	1425.548	0.538
500RG–2DCNN	4.952	1.519	0.154	3096.910	0.541	3.974	1.157	−0.686	1533.762	0.525
10RG-LSTM	4.362	1.422	0.269	2233.650	0.557	3.788	0.826	0.004	994.975	0.526
100RG-LSTM	4.094	1.234	0.145	2459.210	0.522	3.610	0.802	−0.319	974.822	0.519
500RG-LSTM	3.989	1.273	0.110	1880.110	0.522	3.485	0.596	−0.335	379.522	0.524
10LG–3DCNN	4.215	1.348	0.070	3306.320	0.534	3.865	1.076	0.318	1499.332	0.547
100LG–3DCNN	4.877	1.470	0.563	4386.740	0.542	4.116	1.248	0.346	2032.425	0.537
500LG–3DCNN	3.374	1.168	−2.575	704.500	<u>0.560</u>	3.646	1.030	−1.035	750.508	0.545
10LG–2DCNN	4.370	0.549	0.395	−2888.660	<u>0.532</u>	3.810	0.520	0.290	−922.435	0.535
100LG–2DCNN	4.443	2.046	<u>0.624</u>	4654.750	0.533	3.807	1.229	0.397	1784.915	0.534
500LG–2DCNN	4.629	1.146	−0.478	1093.650	0.536	4.253	0.967	−0.316	806.525	0.525
10LG-LSTM	3.984	1.157	−0.044	−156.520	0.531	3.582	0.794	−0.029	−71.032	0.521
100LG-LSTM	4.327	1.442	0.004	4389.920	0.533	3.685	0.892	0.106	1715.122	0.528
500LG-LSTM	4.288	1.370	0.308	3435.520	0.509	3.596	1.011	0.298	1238.022	0.521
10TG–3DCNN	4.739	1.739	0.560	5404.180	0.544	3.223	1.644	0.769	2124.445	0.560
100TG–3DCNN	4.254	1.745	0.312	2615.570	0.532	3.265	0.988	0.464	1189.048	0.528
500TG–3DCNN	4.441	1.280	0.242	1458.900	0.549	3.287	1.003	0.312	839.975	0.543
10TG–2DCNN	3.790	0.625	0.172	−901.210	0.537	2.794	0.430	0.408	−573.745	0.551
100TG–2DCNN	3.969	1.039	0.291	291.930	0.554	3.442	0.835	0.326	155.270	<u>0.557</u>
500TG–2DCNN	3.747	0.789	0.343	−643.220	0.571	<u>2.827</u>	0.419	0.366	−333.225	0.555
10TG-LSTM	<u>3.681</u>	1.257	0.265	2851.330	0.513	<u>2.805</u>	0.622	0.320	1016.572	0.514
100TG-LSTM	3.890	1.455	0.539	3728.170	0.552	2.939	0.972	0.467	1472.445	0.542
500TG-LSTM	3.973	1.478	0.582	3808.950	0.571	2.936	0.797	0.444	1123.458	0.535
Buy & Hold	0.738	0.141	−1.464	15.522	0.532	0.738	0.141	−1.464	15.522	0.532

Note: The bolded sections represent the best performance across each metric, and the underlined sections indicate the second-best performance across each metric.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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